

Lecture 6 - Column Spaces & Nullspaces

- Vector space $\rightarrow$ All reachable vectors by taking linear combination of lets say u & w .

eg $\mathbb{R}^3 \rightarrow$ vector space

$\{0\} \rightarrow$ vector space

$\{[1, 1, 1]\} \rightarrow$ a point in $\mathbb{R}^3$ is not
 $\{ \text{plane in } \mathbb{R}^3 \} \rightarrow$ vectorspace $\checkmark$
through 0 subspace in $\mathbb{R}^3$

$\Rightarrow$ All hyperplanes and lower dimensional planes passing through origin are subspaces $\checkmark$ in $\mathbb{R}^n$.

S_1 & S_2 are 2 subspaces

- $S_1 \cup S_2$ is a subspace iff S_1 is a subspace of S_2 or vice versa
- $S_1 \cap S_2$ is a subspace

Column space $\rightarrow$ Span of linear combinations of columns of matrix A !

$$\begin{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} & \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} \end{bmatrix}$$

$v_1 \quad v_2 \quad v_3$

$$\rightarrow \boxed{a v_1 + b v_2 + c v_3}$$

$\downarrow$
Subspace in $\mathbb{R}^4$

- $C(A) = aV_1 + aV_2 + aV_3$

in given example $C(A)$ is a subspace, specifically a 2 dimensional subspace since
 $3^{rd} \text{ col} = 1^{st} \text{ col} + 2^{nd} \text{ col}$

→ $Ax = b$ solution only exists if b vector exists in the column space of A .

Nullspace

Nullspace of A is defined as all x that solves $\underline{Ax = 0}$

→ This given example has a subspace nullspace in $\mathbb{R}^3$ $[x_1, x_2, x_3]$

→ Given example's nullspace is $\lambda [1, 1, -1] \quad \forall \lambda \in \mathbb{R}$

$$N(A) = \lambda \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

→ This nullspace is a 1-d line in $\mathbb{R}^3$.

→ Solutions to $Ax = 0$ always give a subspace ...

$$Ax_1 = 0 \quad + \quad Ax_2 = 0$$

$$A(x_1 + x_2) = 0$$

↳ linear combination is also in subspace

$\Rightarrow$ Suppose the right hand side is not zero.

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is a possible solution.

$x = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ is also a solution.

$$\lambda A x_1 = \lambda b$$

$$A x_2 = b$$

$$A(\lambda x_1 + x_2) = \underbrace{b(\lambda+1)}_{\rightarrow \text{no subspace!}}$$